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Thesis/Dissertation Template in L^AT_EX

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本論文係洪睿帆君（R05546030）在國立臺灣大學工業工程學研究所完成之碩士學位論文，於民國 109 年 5 月 1 日承下列考試委員審查通過及口試及格，特此證明

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你總是狠狠吃定我，吃定我對你，愛的那麼多多少次打勾勾，多少次我照做

你說，牽手可是，會把女生，的心牽走，我毫不猶豫的緊握我說，能不能夠，能不能夠，讓時間停留，在你還沒放手的時候

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關鍵字： LaTeX、中文、論文、模板



Abstract

Abstract

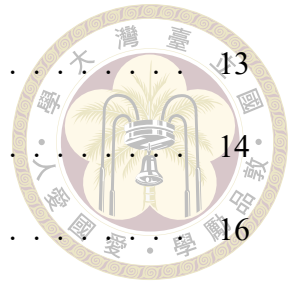
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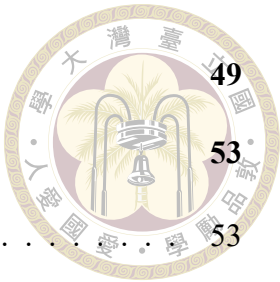
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Denotation



Chapter 1 Introduction

As the demand of high speed and low latency communication networks, future wireless systems must support higher spectral efficiency, flexible resource sharing, and massive connectivity. To increase the channel capacity and reliability, Multiple Access(MA) techniques become more important. However, Orthogonal Multiple Access (OMA) avoids inter-user interference by assigning disjoint resources, its user capacity is fundamentally limited by the number of orthogonal resource units[4]. In particular, beyond 5G and toward 6G, Non-Orthogonal Multiple Access (NOMA) is regarded as a potential multiple access technique to support the growing demands for massive connectivity and diverse service requirements.[3].



Chapter 2 Single resource systems

This chapter studies uplink NOMA with single resource system, where PD-NOMA with soft SIC and GDMA[8] are discussed. Last, we clarify the performance gap between GDMA detection and soft SIC detection under different large scale fading coefficients.

2.1 Single resource system model

Consider an uplink system where perfect timing synchronization is assumed. U users transmit simultaneously over one shared resource, which may be a frequency band, time slot, or spreading signature. The received signal at the BS is

$$y = \sum_{k=1}^U \sqrt{P_k \beta_k} h_k x_k + n, \quad (2.1)$$

where P_k is the transmit power of user k , β_k is the large-scale fading coefficient of user k , h_k is the small-scale Rayleigh flat-fading coefficient of user k , x_k is transmission symbol of user k and $n \sim \mathcal{CN}(0, N_0)$ is complex AWGN. Equivalently, the noise variance per dimension is $\sigma^2 = N_0/2$. In this work, for simplicity, $P_k = 1$ is assumed unless otherwise stated.

2.2 Detection principle for PD NOMA : soft SIC



In PD-NOMA, users are mainly separated by their received-power difference. The instantaneous received-power metric of user k can be written as

$$\rho_k = P_k \beta_k |h_k|^2 \triangleq |g_k|^2. \quad (2.2)$$

The SIC detection order is denoted by

$$\pi = \{\pi(1), \pi(2), \dots, \pi(U)\}, \quad (2.3)$$

where $\pi(i)$ is the user detected at stage i . For power-based ordering, the order is usually determined such that $\rho_{\pi(1)} \geq \rho_{\pi(2)} \geq \dots \geq \rho_{\pi(U)}$. The renumbering of user index is then processed by following the orders. For simplicity, we assume the indexes of users in equation (2.1) have been renumbering. At the first SIC stage, the LLR of x_1 , L_{x_1} , is obtained by regarding signals from other users as approximately Gaussian in equation (2.1) and the soft estimate is denoted as $\hat{s}_1$. At the $(i+1)^{th}$ SIC stage, the LLR of x_{i+1} , $L_{x_{i+1}}$, is obtained by a residual signal $y_{(i+1)}$ which is obtained by subtracting the contribution of the soft estimate $\hat{s}_i$ in the i^{th} stage and regarding signals from other users as Gaussian. Therefore,

$$y_{(1)} = y, \quad (2.4)$$

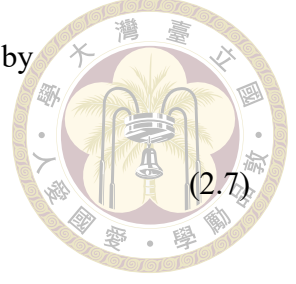
$$y_{(i+1)} = y_{(i)} - g_i \hat{s}_i. \quad (2.5)$$

For BPSK modulation, the log likelihood ratio (LLR) of user x_i can be approximated as

$$L_{x_i} = \log \frac{p(y_{(i)} | x_{\pi(i)} = +1)}{p(y_{(i)} | x_{\pi(i)} = -1)} = \frac{4|g_i^* y_{(i)}|^2}{\sigma_{\text{eff},(i)}^2}. \quad (2.6)$$

The soft estimate of the detected symbol given $y_{(i)}$ is expressed [20] by

$$\hat{s}_i = \mathbb{E}[x_i | L_{x_i}] = \tanh\left(\frac{L_{x_i}}{2}\right). \quad (2.7)$$



and the conditional variance is obtained [20] as

$$\sigma_{\hat{s}_i}^2 = \mathbb{E}[x_i^2 | L_{x_i}] = 1 - \tanh^2\left(\frac{L_{x_i}}{2}\right). \quad (2.8)$$

Therefore, the effective noise variance is

$$\sigma_{\text{eff},(i)}^2 = 2\sigma^2 + \sum_{j=i+1}^U |g_j|^2 + \sum_{j=1}^{i-1} \sigma_{\hat{s}_j}^2. \quad (2.9)$$

This procedure is repeated until all users are detected.

2.3 Detection principle for GDMA

In GDMA, users are distinguished by their distinct channel gains. Unlike PD-NOMA, which relies mainly on received-power differences, GDMA uses both the magnitude and phase of g_k .

l	x_1	x_2	s_l
0	1	1	$\sqrt{\beta_1}h_1 + \sqrt{\beta_2}h_2$
1	1	-1	$\sqrt{\beta_1}h_1 - \sqrt{\beta_2}h_2$
2	-1	1	$-\sqrt{\beta_1}h_1 + \sqrt{\beta_2}h_2$
3	-1	-1	$-\sqrt{\beta_1}h_1 - \sqrt{\beta_2}h_2$

Table 2.1: Mapping rules for superimposed signal in the case of $m = 1$ and $U = 2$

At the receiver, the superimposed signal is first used to calculate the *a posteriori* probabilities (APPs) of all possible combinations of multiuser symbols, and then the LLR

of each user is then obtained accordingly[8]. There are 2^{MU} possible superimposed signal levels, denoted by the set

$$\mathcal{S} = \{s_0, s_1, \dots, s_{2^{MU}-1}\}, \quad (2.10)$$

where M is the modulation order and $s_l \in \mathcal{S}$ contributes to the signal part in (2.1). Take BPSK modulation ($M = 1$) as example, the superimposed signal

$$\begin{aligned} s_l &= (-1)^{l_2[1]} \cdot \sqrt{\beta_1} h_1 + (-1)^{l_2[2]} \cdot \sqrt{\beta_2} h_2 + \dots + (-1)^{l_2[U]} \cdot \sqrt{\beta_U} h_U \\ &= \sum_{k=1}^U (-1)^{l_2[k]} \cdot \sqrt{\beta_k} h_k, \quad l = 0, 1, \dots, 2^U - 1, \end{aligned} \quad (2.11)$$

where $l_2[k]$ denotes the transmission bit of user k in the binary representation.

In detail:

$$\begin{aligned} l = 0 &\rightarrow (l_2[1], l_2[2], \dots, l_2[U]) = (0, 0, \dots, 0) \\ l = 1 &\rightarrow (l_2[1], l_2[2], \dots, l_2[U]) = (0, 0, \dots, 1) \\ &\vdots \\ l = 2^U - 1 &\rightarrow (l_2[1], l_2[2], \dots, l_2[U]) = (1, 1, \dots, 1). \end{aligned} \quad (2.12)$$

The superimposed signal levels s_l for the case $U = 2$ are listed in Table 2.1, where $l \in \{0, 1, 2, 3\}$.

The receiver computes the *a posteriori* probabilities (APPs) of each possible superimposed signal $s_l \in \mathcal{S}$ by

$$P(s_l|y) = P(y|s_l) \cdot \frac{P(s = s_l)}{P(y)}, \quad (2.13)$$

while

$$P(y|s_l) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{|y - s_l|^2}{2\sigma^2}\right). \quad (2.14)$$

Assume the signals in $\mathcal{S}$ are equally propable, the APPs can be written as

$$P(s_l|y) \propto \exp\left(-\frac{|y-s_l|^2}{2\sigma^2}\right). \quad (2.15)$$



The log-likely ratio (LLR) of user k is then calculated by

$$L_{x_k} = \log \frac{\sum_{x_k=+1} \exp\left(-\frac{|y-s_l|^2}{2\sigma^2}\right)}{\sum_{x_k=-1} \exp\left(-\frac{|y-s_l|^2}{2\sigma^2}\right)}. \quad (2.16)$$

2.4 Simulations of PD NOMA with soft SIC and GDMA

The single-resource simulations use BPSK modulation for each user over independent Rayleigh flat-fading channels. Figure 2.1, 2.2 and 2.3 show the BER performance of two users with different large-scale fading coefficients (β_1, β_2) . As the large-scale fading gap becomes larger, the soft SIC detection converges toward the performance of the GDMA detection, indicating that the received-power disparity provides a clearer detection order and reduces the ambiguity of early cancellation stages. However, it comes at the severe cost of user fairness[5].

Figure 2.4 shows the BER performance of three users with large-scale fading coefficients $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$. The three-user result further clarifies the limitation of soft SIC detection. As the number of users increases, soft SIC becomes more susceptible to error propagation because each cancellation stage depends on the reliability of the previous soft estimates. In contrast, GDMA evaluates the multiuser signal constellation as a whole and can therefore maintain favorable BER performance as the number of users increases[15]. This performance advantage comes with an exponential growth cost with

U .

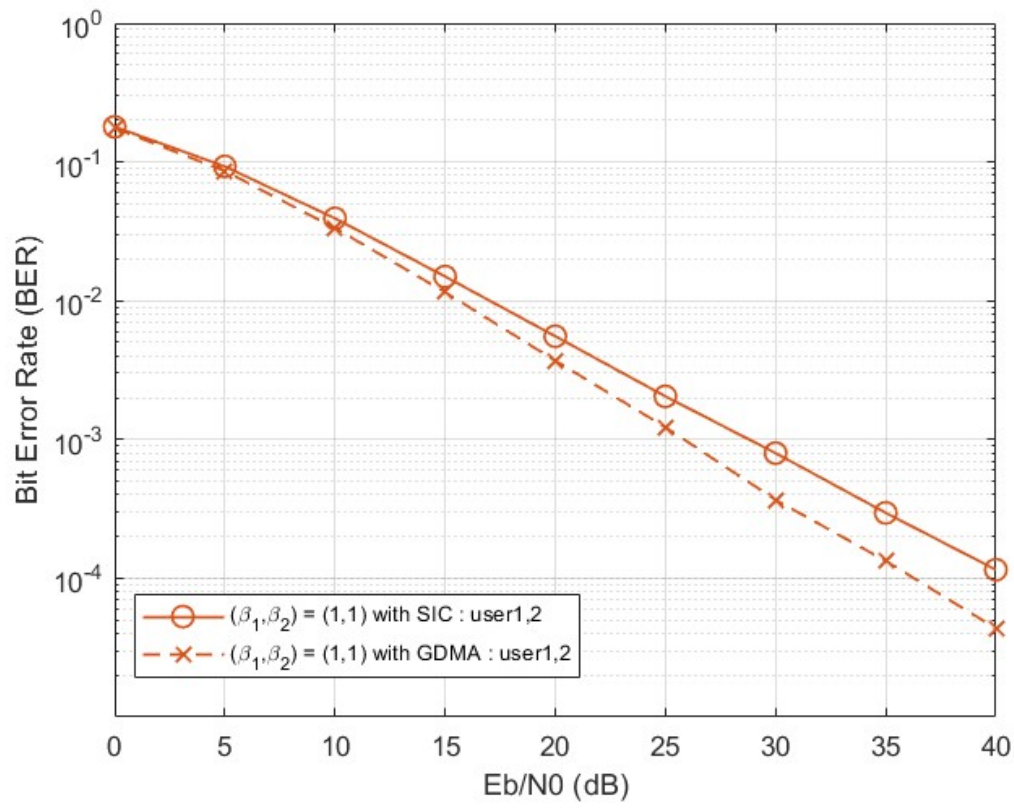
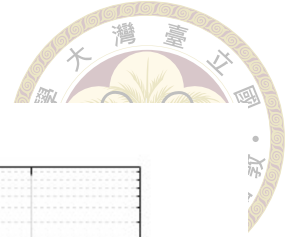


Figure 2.1: Users' BERs for single resource systems of $(\beta_1, \beta_2) = (1, 1)$.

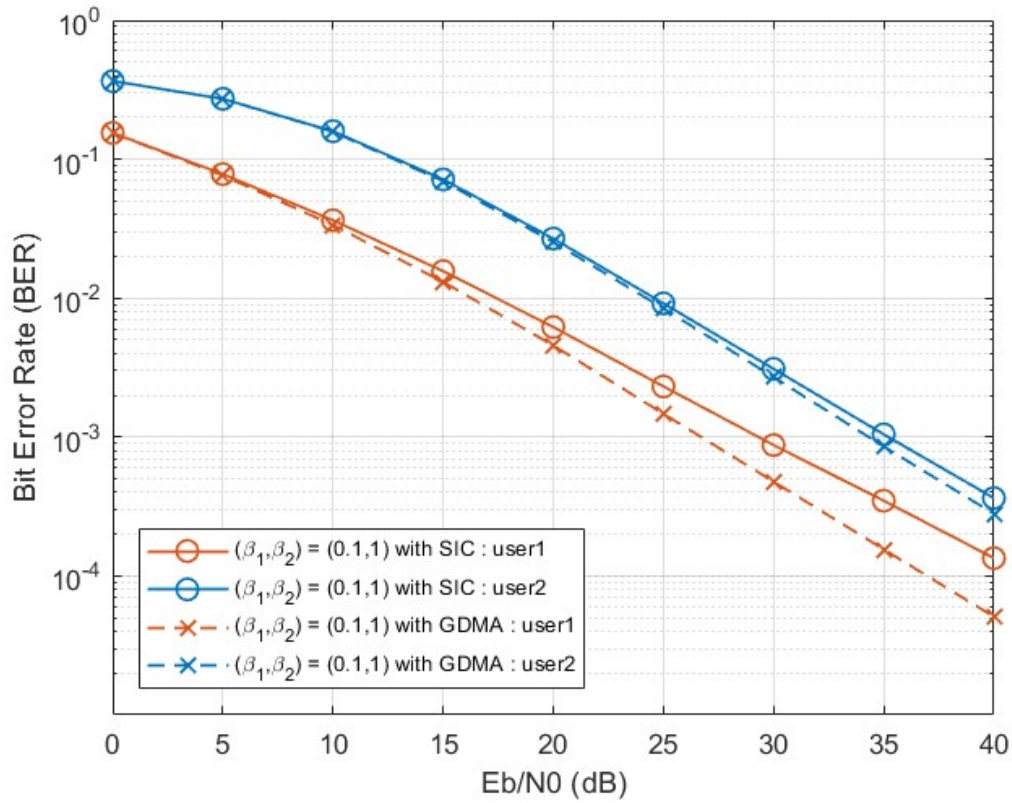
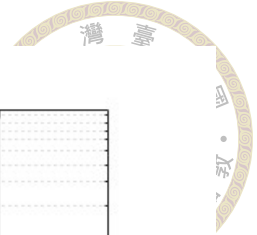


Figure 2.2: Users' BERs for single resource systems of $(\beta_1, \beta_2) = (1, 0.1)$.

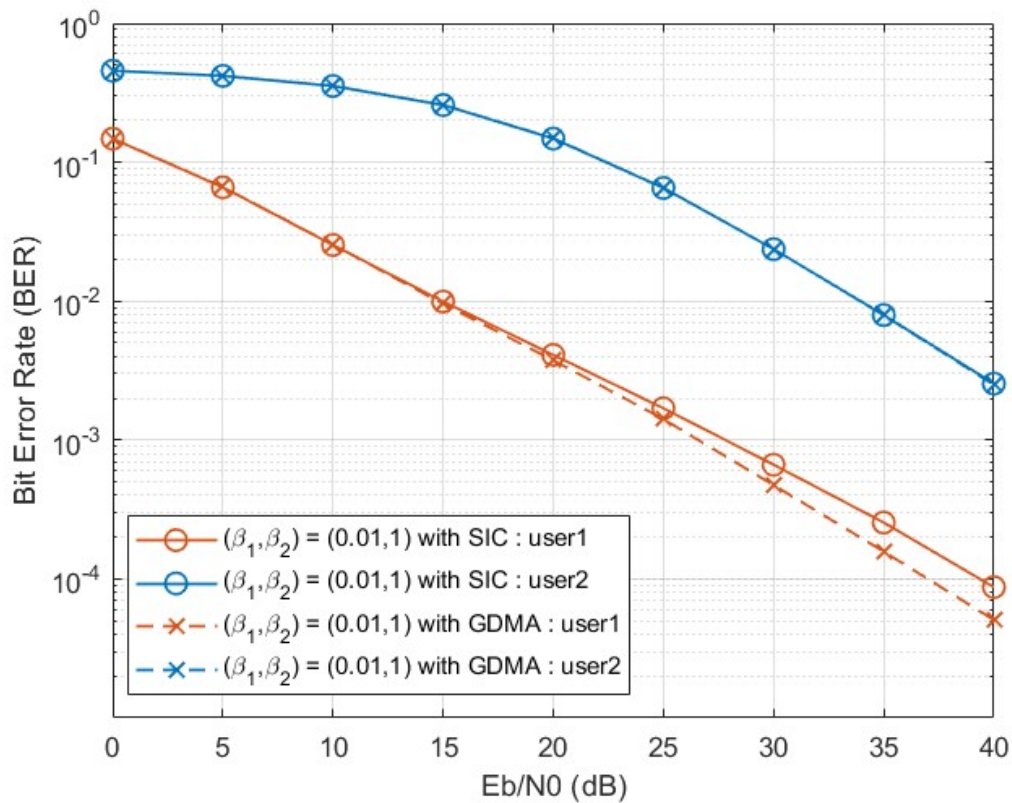


Figure 2.3: Users' BERs for single resource systems of $(\beta_1, \beta_2) = (1, 0.01)$.

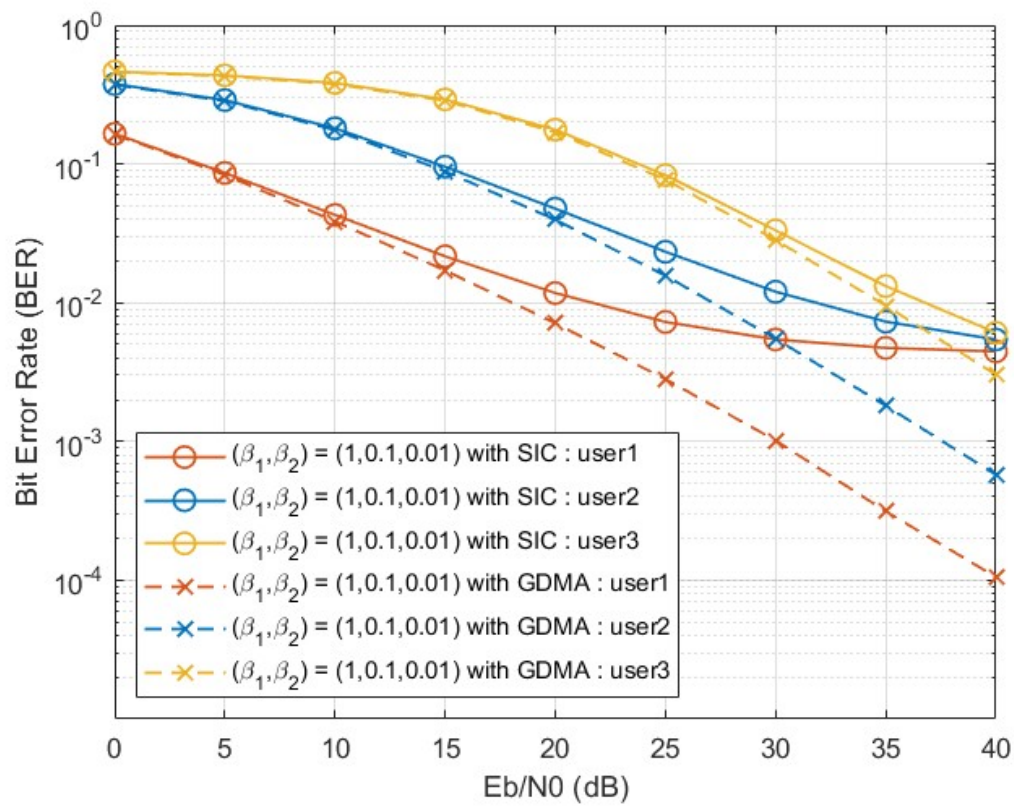


Figure 2.4: Users' BERs for single resource systems of $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$.



Chapter 3 Multiple resource NOMA systems

In this chapter, we extend single-resource PD-NOMA to a multiple-resource system, where several shared-resource observations are used to improve multiuser separation ability. Since single-resource PD-NOMA mainly relies on received-power differences, the multiple-resource model allows the BS to further exploit resource-domain channel-vector information. For multi-resource PD-NOMA, MMSE-SIC detection is used, post-SINR ordering and amplitude-based ordering are considered as two detection-ordering methods, and the performance is compared with HDS-CDMA with ML detection [13].

3.1 Multi-resource PD-NOMA system model

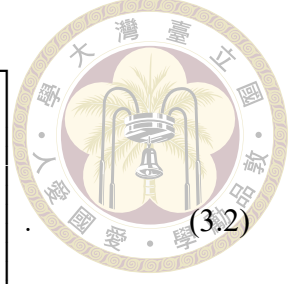
Multi-resource PD-NOMA refers to an architecture in which each user can transmit over multiple shared resources. At one symbol time, the received vector is

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}, \quad (3.1)$$

where $\mathbf{y} \in \mathbb{C}^{R \times 1}$ are the received signals over R resources, $\mathbf{x} \in \mathbb{C}^{U \times 1}$ are the transmitted signals of U users and $\mathbf{n} \in \mathbb{C}^{R \times 1}$ denotes complex AWGN. The channel matrix $\mathbf{H} \in$

$\mathbb{C}^{R \times U}$ is

$$\mathbf{H} = \begin{bmatrix} \sqrt{P_1\beta_1}h_{1,1} & \sqrt{P_2\beta_2}h_{1,2} & \cdots & \sqrt{P_U\beta_U}h_{1,U} \\ \sqrt{P_1\beta_1}h_{2,1} & \sqrt{P_2\beta_2}h_{2,2} & \cdots & \sqrt{P_U\beta_U}h_{2,U} \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{P_1\beta_1}h_{R,1} & \sqrt{P_2\beta_2}h_{R,2} & \cdots & \sqrt{P_U\beta_U}h_{R,U} \end{bmatrix}. \quad (3.2)$$



The k th column $\mathbf{h}_k$ represents the resource-domain channel vector of user k . For the representative two-resource and three-user case, the model is

$$\begin{bmatrix} y_A \\ y_B \end{bmatrix} = \begin{bmatrix} h_{A,1} & h_{A,2} & h_{A,3} \\ h_{B,1} & h_{B,2} & h_{B,3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} n_A \\ n_B \end{bmatrix}, \quad (3.3)$$

where the large-scale fading and transmit-power factors are absorbed into each $h_{r,k}$.

We employ MMSE-SIC detection for equation (3.1). The resulting filter structure is similar to those widely used in multi-antenna systems [6, 10] and pattern division multiple access (PDMA) receivers [9]. In multi-antenna reception, user separation mainly comes from spatial observations, while in PDMA it is further assisted by user-specific pattern mapping across resources. In contrast, the considered multi-resource PD-NOMA system does not adopt sparse signatures or pattern design[17]; instead, it relies on the channel vectors across shared resources to improve user separation reliability and overall performance.

3.2 MMSE-SIC Detection

The MMSE-SIC receiver reduces complexity by successively detecting users after linear MMSE filtering. The filter first suppresses the interference components that are

linearly separable from the desired user's channel information, and the SIC stage then removes the detected user's contribution from the residual vector. Compared with single-resource soft SIC, this receiver can exploit both magnitude and phase differences across resources. The main quantities needed after filtering are the desired effective channel gain, the residual interference-plus-noise variance, the LLR, and the soft estimate used for cancellation. For the multi-resource model of equation (3.1), consider user k be detected. The MMSE filter is designed by following [9, Eqs. (6), (7)] with the coefficient

$$\mathbf{w}_k^H = \mathbf{h}_k^H (\mathbf{H} \mathbf{H}^H + 2\sigma^2 \mathbf{I})^{-1}, \quad (3.4)$$

The filter output of user k can be formulated as

$$z_k = \mathbf{w}_k^H \mathbf{y} = \mathbf{w}_k^H \mathbf{h}_k x_k + \sum_{j \neq k} \mathbf{w}_k^H \mathbf{h}_j x_j + \mathbf{w}_k^H \mathbf{n}. \quad (3.5)$$

In equation (3.5), we define the effective channel gain:

$$a_k = \mathbf{w}_k^H \mathbf{h}_k, \quad (3.6)$$

and the effective residual interference-plus-noise variance:

$$\sigma_{k,\text{eff}}^2 = \sum_{j \neq k} |\mathbf{w}_k^H \mathbf{h}_j|^2 + 2\sigma^2 \|\mathbf{w}_k^H\|^2. \quad (3.7)$$

3.2.1 Detection Ordering



To determine successive detection orders in the SIC process, two ordering strategies are considered. The first is post-SINR ordering [10]. The post-filtering SINR of user k is

$$\gamma_k^{\text{post}} = \frac{|w_k^H h_k|^2}{\sum_{j \neq k} |w_k^H h_j|^2 + 2\sigma^2 \|w_k\|^2}. \quad (3.8)$$

The receiver detects the user with the largest post-SINR first. This criterion uses both the magnitude and phase of the channel information, and it reflects the actual quality of each user after MMSE filtering, which are generally preferred in reducing SIC error propagation.

The second strategy is amplitude-based ordering. Users are sorted according to a channel magnitude or received-power metric such as

$$\gamma_k^{\text{amp}} = \|h_k\|^2. \quad (3.9)$$

This ordering is simpler because it relies only on channel magnitudes, but ignores how MMSE filtering suppresses multiuser interference. After determining the SIC orders and then re-numbering the user indexes, we assume

$$\gamma_1 \geq \dots \geq \gamma_U \quad (3.10)$$

for simplicity. This renumbering is applied for either one of the ordering strategies and simplifies the superscript.

3.2.2 SIC Procedure



After the order is decided, users are renumbered as $1, 2, \dots, U$. At stage i , the residual vector is $\mathbf{y}^{(i)}$ and the corresponding channel matrix is

$$\mathbf{H}_{(i)} = [\mathbf{h}_i, \mathbf{h}_{i+1}, \dots, \mathbf{h}_U]. \quad (3.11)$$

The MMSE filter output and the effective variance of the residual interferences and noise can be derived as

$$z_i = \mathbf{w}_i^H \mathbf{y}^{(i)} = \mathbf{w}_i^H \mathbf{h}_i x_i + \sum_{j>i} \mathbf{w}_i^H \mathbf{h}_j x_j + \mathbf{w}_i^H \mathbf{n}. \quad (3.12)$$

$$\sigma_{i,\text{eff}}^2 = \sum_{j>i} |\mathbf{w}_i^H \mathbf{h}_j|^2 + 2\sigma^2 \|\mathbf{w}_i^H\|^2. \quad (3.13)$$

After MMSE filtering, the output can be approximated as an equivalent scalar channel

$$z_i = a_i x_i + \eta_i, \quad a_i = \mathbf{w}_i^H \mathbf{h}_i, \quad \eta_i \sim \mathcal{CN}(0, \sigma_{i,\text{eff}}^2). \quad (3.14)$$

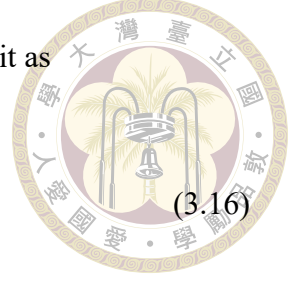
For a general bit-labeled modulation alphabet $\mathcal{S}$, let $b_m(s) \in \{0, 1\}$ denote the m th label bit of symbol s , and define

$$\mathcal{S}_{m,b} = \{s \in \mathcal{S} : b_m(s) = b\}, \quad b \in \{0, 1\}. \quad (3.15)$$

In this chapter, the transmitted coded bits are assumed to be independent and equally probable. Therefore, the prior term cancels in the likelihood dfuncyion. Under the scalar Gaussian approximation in equation(3.14), the standard likelihood-marginalization form

of the soft-output demapper [20] gives the detector LLR of the m th bit as

$$L_{x_i,m} = \log \frac{\sum_{s \in \mathcal{S}_{m,0}} \exp \left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2} \right)}{\sum_{s \in \mathcal{S}_{m,1}} \exp \left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2} \right)}, \quad (3.16)$$



For BPSK modulation, $\mathcal{S} = \{+1, -1\}$ with $0 \mapsto +1$ and $1 \mapsto -1$, and equation (3.16) becomes

$$L_{x_i} = \frac{4|a_i^* z_i|^2}{\sigma_{i,\text{eff}}^2}. \quad (3.17)$$

The corresponding soft symbol estimate is the posterior mean obtained from the same equally likely symbol prior [20]:

$$\hat{s}_i = \mathbb{E}[x_i | z_i] = \frac{\sum_{s \in \mathcal{S}} s \exp \left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2} \right)}{\sum_{s \in \mathcal{S}} \exp \left(-\frac{|z_i - a_i s|^2}{\sigma_{i,\text{eff}}^2} \right)}. \quad (3.18)$$

This posterior mean reduces to $\hat{s}_i = \tanh(L_i/2)$ if BPSK modulation is used. The expression in equation (3.18) is therefore a detector-output soft estimate for the uncoded receiver when all transmitted bits are equally likely. The soft estimate is then subtracted from the received vector:

$$\mathbf{y}_{(i+1)} = \mathbf{y}_{(i)} - \mathbf{h}_i \hat{s}_i. \quad (3.19)$$

The process then goes back to re-calculate the coefficient $\mathbf{w}_{i+1}^H$ by using equation (3.4) with $\mathbf{H}_{(i+1)}$. The LLR of x_{i+1} and its soft estimate $\hat{s}_{i+1}$ can also be obtained by equation (3.16) and (3.18). This procedure is repeated until all users are detected.

If $\hat{s}_i$ is unreliable, the subtraction leaves a residual of cancellation in $\mathbf{y}_{(i+1)}$. This residual is the main source of error propagation in MMSE-SIC, and motivates the variance-aware coded receiver discussed in Chapter 4.



3.3 Uncoded Simulation Results

The main uncoded multi-resource setting uses BPSK transmission, $R = 2$ resources, $U = 3$ users, independent Rayleigh flat fading, and perfect CSI are assumed. The MMSE-SIC detector is compared with HDS-CDMA with ML detection[13]. We consider two large-scale fading cases: $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$ and $(\beta_1, \beta_2, \beta_3) = (1, 1, 1)$. The average transmit energy per bit per user is normalized by the number of resources.

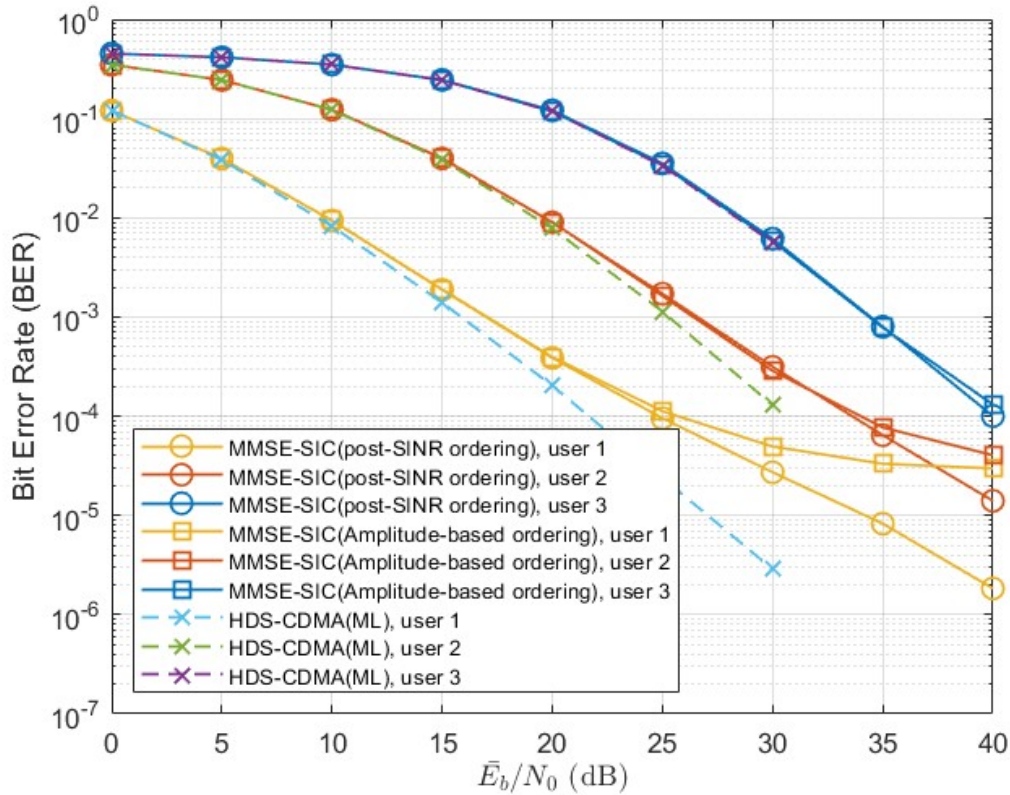


Figure 3.1: Multi-resource system. BER of MMSE-SIC and HDS-CDMA with ML detection with $R = 2$, $U = 3$ and $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$.

The results in figure. 3.1 and 3.2 show that HDS-CDMA with ML detection achieves better uncoded BER than MMSE-SIC in both large-scale fading cases. For the unequal- β_k case in figure. 3.1, HDS-CDMA performs about 3 dB better than MMSE-SIC at BER = 10^{-4} for user 1. For the equal- β_k case, HDS-CDMA performs about 7 dB better than

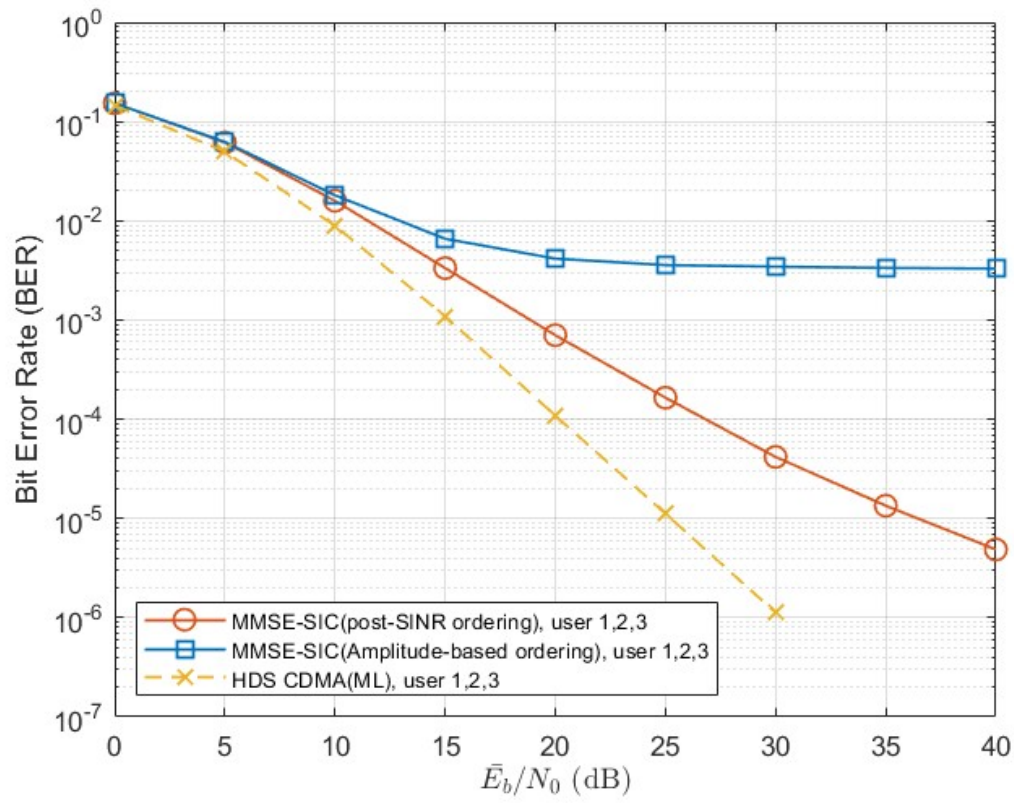


Figure 3.2: Multi-resource system. BER of MMSE-SIC and HDS-CDMA with ML detection with $R = 2$, $U = 3$ and $(\beta_1, \beta_2, \beta_3) = (1, 1, 1)$.

MMSE-SIC at $\text{BER} = 10^{-4}$. Therefore, the performance loss of MMSE-SIC is much more significant when the users have similar average received powers.

The smaller BER gap of two detector in the unequal-fading case is caused by the large-scale fading disparity. Since the users have clearly separated average received powers, the first SIC stage is more reliable and the cancellation error is less likely to dominate the following stages. In particular, for the user with the smallest average received power, MMSE-SIC is close to HDS-CDMA within the simulated SNR range. However, HDS-CDMA still provides the best overall BER performance because of its joint detection ability.

The ordering comparison further explains the behavior of MMSE-SIC. In the unequal- β_k case, amplitude-based ordering only causes a slight degradation around 25 dB for user 1. In the equal- β_k case, however, amplitude-based ordering degrades heavily because users do not have a clear large-scale power separation, indicating that post-SINR ordering is more robust because it measures the quality after MMSE filtering, whereas amplitude-based ordering only uses the channel magnitude before filtering.

These results indicate that multi-resource PD-NOMA improves user separation compared with single-resource soft SIC, but HDS-CDMA with ML detection remains the stronger uncoded benchmark. The remaining performance gap motivates the next chapter. Chapter 4 introduces channel coding and iterative detection and decoder (IDD) so that the reliability of the decoder can be used to reduce the propagation of SIC errors.



Chapter 4 Coded Multi-Resource Systems with IDD

This chapter introduces channel coding and iterative detection and decoding (IDD) into the multi-resource receiver. Motivated by the error-propagation problem observed in uncoded SIC receivers, the detector exchanges soft information with the FEC decoder and refines interference cancellation through iterations based on a priori information update. The block error rate (BLER) performance gap of MMSE-SIC-IDD, MMSE-PIC-IDD and MAP-IDD are clarified in coded multi-resource uplink NOMA system.

4.1 Introduction of IDD structure

For coded systems, the detector is typically embedded in an iterative detection and decoding (IDD) receiver. In this structure, soft information is exchanged between the detector and the FEC decoder [11, 21, 23]. The detector produces bit-level LLRs for the coded bits, while the decoder uses the code constraints to refine the reliability information and returns updated soft information to the detector, which is the a priori information of next detector calculation. In our simulations, the IDD receiver is implemented with a maximum number of iterations $t_{\max}$, and the final decoder output is used to make hard

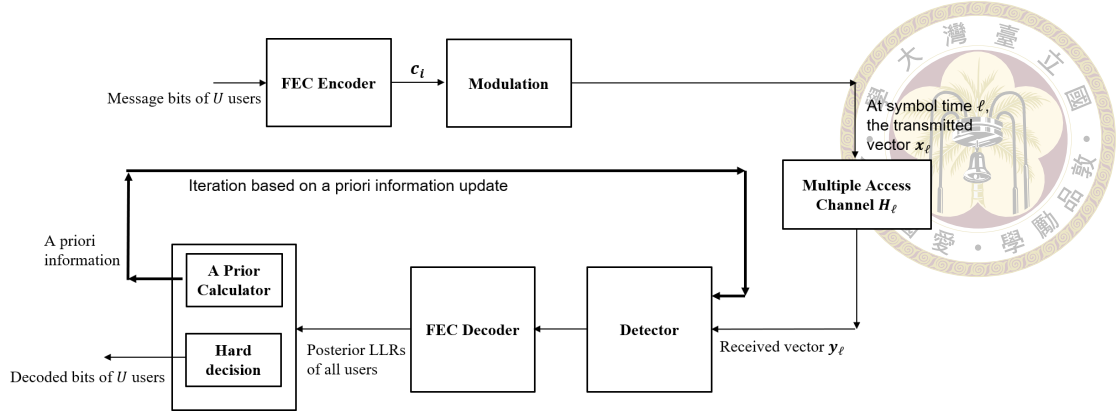


Figure 4.1: Coded NOMA system with IDD receiver

decisions on the transmitted bits.

4.2 Coded Multi-resource system model

In this chapter, different users are allowed to employ different modulation schemes and different channel code lengths. Let c_i denote user i 's transmitted codeword of length $N_{c,i}$

$$\mathbf{c}_i = [c_{i,1}, c_{i,2}, \dots, c_{i,N_{c,i}}]^T, \quad (4.1)$$

After modulation, the codeword is grouped into N_s modulation symbols. At symbol time ℓ , the transmitted vector is

$$\mathbf{x}_\ell = [x_{1,\ell}, x_{2,\ell}, \dots, x_{U,\ell}]^T, \quad (4.2)$$

Refer to equation (3.1), the received vector is

$$\mathbf{y}_\ell = \mathbf{H}_\ell \mathbf{x}_\ell + \mathbf{n}_\ell. \quad (4.3)$$

In the block-fading simulations, $\mathbf{H}_1 = \mathbf{H}_2 = \dots = \mathbf{H}_{N_s}$. For simplicity, we omit the subscript ℓ , which denotes the process is dealing with $\mathbf{y}_\ell$. The overall block diagram of

coded NOMA system with IDD receiver is shown in figure. 4.1.



4.3 MMSE-SIC-IDD

In this section, we introduce the procedure of the MMSE-SIC-IDD receiver. The key feature of the structure is that interference cancellation in the SIC process is performed after channel decoding rather than immediately after detection.

4.3.1 Receiver Procedure

Following a similar procedure in section 3.2.1, the detection order can be obtained. For post-SINR ordering method, the additional $\bar{v}_i^{(t)}$ is considered and will be discussed in section 4.3.3. Assume the users are re-numbered from 1 to U according to this order.

For user i , the detector first computes the bit-level LLRs L_{c_i} . The summation of these LLRs and a priori information are then passed to the FEC decoder of user i . After decoding, the decoder produces the a posteriori LLRs $L_{c_i}^d$, which are used to generate the decoded soft symbol estimate $\hat{s}_i^d$. This decoded soft symbol estimate is then fed back to the MMSE-SIC detector for the next SIC stage, i.e., user $i + 1$, where it is used to subtract the interference caused by user i . Intuitively, performing cancellation after decoding can improve the reliability of the cancelled signal and thus reduce error propagation in the SIC process.

Due to the successive decoding procedure, the extrinsic information of each user is collected asynchronously. Specifically, after decoding user i , the extrinsic LLRs are

obtained as

$$\mathbf{L}_{c_i}^e = \mathbf{L}_{c_i}^d - \mathbf{L}_{c_i}^{in}, \quad (4.4)$$

where $\mathbf{L}_{c_i}^{in}$ is the input of decoder. In this way, the decoded extrinsic information of all users is provided as a priori information to the MMSE-SIC detector in the following IDD iteration.

4.3.2 Soft Information for a priori update

Let $\mathcal{S}_i$ denote the modulation alphabet of user i , and let $M_i = \log_2 |\mathcal{S}_i|$ be the number of coded bits per modulation symbol. For a symbol $s \in \mathcal{S}_i$, its binary label is written as

$$\mathbf{b}_i(s) = [b_{i,0}(s), b_{i,1}(s), \dots, b_{i,M_i-1}(s)]. \quad (4.5)$$

Let $L_{c_i,\ell,m}^{(t)}$ denote the a priori LLR of the m th coded bit in the ℓ th modulation symbol of user i at a priori update iteration t , with

$$L_{c_i,\ell,m}^{(t)} = \log \frac{P^{a,(t)}(b_{i,\ell,m} = 0)}{P^{a,(t)}(b_{i,\ell,m} = 1)}, \quad L_{c_i,\ell,m}^{(0)} = 0. \quad (4.6)$$

The corresponding bit probability is

$$P^{a,(t)}(b_{i,\ell,m} = b) = \frac{1}{1 + \exp \left[-(1 - 2b)L_{c_i,\ell,m}^{(t)} \right]}, \quad b \in \{0, 1\}. \quad (4.7)$$

Assuming the interleaved coded bits are represented by independent bit priors at the demapper input, the a priori symbol probability is the product of its corresponding a priori bit probability[22], which is written as

$$P_{i,\ell}^{a,(t)}(s) = \prod_{m=0}^{M_i-1} P^{a,(t)}(b_{i,\ell,m} = b_{i,m}(s)). \quad (4.8)$$



The soft symbol mean used for a priori update before detection is calculated according to the a priori symbol probability[20], that is,

$$\mu_{i,\ell}^{(t)} = \mathbb{E}[x_{i,\ell} \mid \mathbf{L}_{c_{i,\ell}}^{(t)}] = \frac{\sum_{s \in \mathcal{S}_i} s P_{i,\ell}^{a,(t)}(s)}{\sum_{s \in \mathcal{S}_i} P_{i,\ell}^{a,(t)}(s)}, \quad (4.9)$$

and the corresponding symbol variance is

$$\sigma_{i,\ell}^{2(t)} = \frac{\sum_{s \in \mathcal{S}_i} |s|^2 P_{i,\ell}^{a,(t)}(s)}{\sum_{s \in \mathcal{S}_i} P_{i,\ell}^{a,(t)}(s)} - \left| \mu_{i,\ell}^{(t)} \right|^2. \quad (4.10)$$

For BPSK modulation with $0 \mapsto +1$ and $1 \mapsto -1$, equation (4.9) and (4.10) reduce to

$$\mu_{i,\ell}^{(t)} = \tanh\left(\frac{L_{c_{i,\ell},0}^{(t)}}{2}\right), \quad \sigma_{i,\ell}^{2(t)} = 1 - \left| \mu_{i,\ell}^{(t)} \right|^2. \quad (4.11)$$

For simplicity, we consider the average variance of user i over the coded block, that is,

$$\bar{v}_i^{(t)} = \frac{1}{N_s} \sum_{\ell=1}^{N_s} \sigma_{i,\ell}^{2(t)}, \quad (4.12)$$

The variance matrix is defined as

$$\mathbf{V} = \text{diag}(\bar{v}_1^{(t)}, \bar{v}_2^{(t)}, \dots, \bar{v}_U^{(t)}), \quad \bar{v}_i^{(t)} > 0. \quad (4.13)$$

When decoder feedback is reliable, $\bar{v}_i^{(t)}$ becomes small, indicating a higher probability of successful decoding for user i and consequently a smaller residual interference. Conversely, when decoder feedback is uncertain, the variance remains large, allowing the receiver to avoid overconfident cancellation.





4.3.3 Variance-Aware MMSE Filtering

When all users' extrinsic information from the FEC decoder has been asynchronously collected, the variance-aware MMSE filter of user k [21, Eqs. (6), (7)] for next a priori based iteration can proceed with the coefficient

$$\tilde{\mathbf{w}}_k^H = \bar{v}_k^{(t)} \mathbf{h}_k^H (\mathbf{H} \mathbf{V} \mathbf{H}^H + 2\sigma^2 \mathbf{I})^{-1}. \quad (4.14)$$

the corresponding post-SINR can be written as

$$\gamma_k^{\text{post}} = \frac{|\tilde{\mathbf{w}}_k^H \mathbf{h}_k|^2}{\sum_{j \neq k} \bar{v}_j^{(t)} |\tilde{\mathbf{w}}_k^H \mathbf{h}_j|^2 + 2\sigma^2 \|\tilde{\mathbf{w}}_k\|^2}. \quad (4.15)$$

Compared with equation (3.4) and (3.8), users with reliable a priori information contribute less uncertainty to the MMSE filtering process and SIC detection ordering.

4.3.4 Decoder-Aided SIC Cancellation

To simplify the expression in the following equations, we omit the subscript ℓ , which denotes that the process is dealing with $\mathbf{y}_\ell$. The procedure goes back to MMSE-SIC stage 1, user 1 can benefit from the a priori information of users $2, \dots, U$ by using

$$\mathbf{y}_{(1)}^{(t)} = \mathbf{y} - \sum_{j=2}^U \mathbf{h}_j \mu_j^{(t)}. \quad (4.16)$$

At SIC stage $i > 1$, the detection of user i benefits from the FEC outputs of users $1, 2, \dots, i-1$ and from the a priori information of users $i+1, \dots, U$. The residual vector

for detecting user i is constructed as

$$\mathbf{y}_{(i)}^{(t)} = \mathbf{y} - \sum_{j=1}^{i-1} \mathbf{h}_j \hat{s}_j^d - \sum_{j=i+1}^U \mathbf{h}_j \mu_j^{(t)}. \quad (4.17)$$



The first summation cancels the users that have already been decoded in the current SIC pass by using the decoder-aided soft symbol estimates $\hat{s}_j^d$. This will be discussed in Section 4.3.5. The second summation cancels the users that have not yet been decoded in the current pass by using the a priori soft symbol mean $\mu_j^{(t)}$. The cancellation term in equation (4.17) differs from that used in the uncoded MMSE-SIC detection. In the uncoded case, the cancelled signal is directly derived from the detector output at the current SIC stage, and the corresponding residual uncertainty is not explicitly considered in MMSE filtering. In the coded IDD receiver, however, users $1, \dots, i-1$ have already been decoded in the current SIC pass and the outputs provide more reliable soft symbol estimates as well as reliability information for quantifying the remaining uncertainty after cancellation. Therefore, the decoder-aided cancellation is modeled together with its residual variance $\bar{v}_c^{can}$, whose detailed calculation is given in Section 4.3.5.

Similar to the uncoded MMSE-SIC procedure in equation (3.11), (3.12), (3.13) and (3.14), the downsized channel matrix and the downsized variance matrix are used after each stage. The filter output for user i is

$$z_i^{(t)} = \tilde{\mathbf{w}}_i^H \mathbf{y}_{(i)}^{(t)}, \quad (4.18)$$

with equivalent scalar gain

$$a_i^{(t)} = \tilde{\mathbf{w}}_i^H \mathbf{h}_i. \quad (4.19)$$

The effective variance of the residual interference and noise with additional residual un-

certainty is

$$\sigma_{i,\text{eff}}^2(t) = \sum_{j>i} \bar{v}_j^{(t)} |\tilde{\mathbf{w}}_i^H \mathbf{h}_j|^2 + \sum_{c<i} \bar{v}_c^{\text{can}} |\tilde{\mathbf{w}}_i^H \mathbf{h}_c|^2 + 2\sigma^2 \|\tilde{\mathbf{w}}_i^H\|^2 \quad (4.20)$$



Unlike the uncoded MMSE-SIC detection, the coded receiver explicitly accounts for the residual uncertainty of decoder-aided cancellation.

4.3.5 LLR calculation and soft estimate

Define the set $\mathcal{S}_{i,m,b}$ is the subset of constellation symbols of user i whose m^{th} bit label is equal to b :

$$\mathcal{S}_{i,m,b} = \{s \in \mathcal{S}_i : b_{i,m}(s) = b\}, \quad b \in \{0, 1\}. \quad (4.21)$$

After MMSE filtering and decoder-aided cancellation, the LLR of detector output is computed from $z_i^{(t)}$, the scalar gain $a_i^{(t)}$, the effective variance $\sigma_{i,\text{eff}}^2(t)$ and the a priori LLRs in equation (4.6). The a priori symbol probability is given by equation (4.8), where the bit probabilities are obtained from $L_{c_i,\ell,m}^{(t)}$ by the LLR-to-probability rule in equation (4.7). The subscript ℓ and superscript (t) are omitted in the following scalar detection equations and LLR expressions respectively for compactness. The detector produces an extrinsic bit-level LLR for the m^{th} bit of user i , $L_{c_i,m}$ [22] by using the a priori information of the other bits in the same modulation symbol, but not the a priori information of the target bit

itself:

$$L_{c_i,m} = \log \frac{\sum_{s \in S_{i,m,0}} \exp \left(-\frac{|z_i^{(t)} - a_i^{(t)} s|^2}{\sigma_{i,\text{eff}}^2(t)} + \sum_{\substack{m'=0 \\ m' \neq m}}^{M_i-1} \log P^{a,(t)}(b_{i,m'} = b_{i,m'}(s)) \right)}{\sum_{s \in S_{i,m,1}} \exp \left(-\frac{|z_i^{(t)} - a_i^{(t)} s|^2}{\sigma_{i,\text{eff}}^2(t)} + \sum_{\substack{m'=0 \\ m' \neq m}}^{M_i-1} \log P^{a,(t)}(b_{i,m'} = b_{i,m'}(s)) \right)}. \quad (4.22)$$

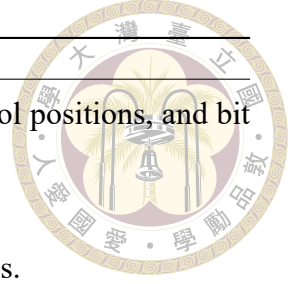
The input of FEC decoder is the sum of the detector extrinsic bit LLR and the target-bit a priori LLR:

$$L_{c_i,m}^{\text{in}} = L_{c_i,m} + L_{c_i,m}. \quad (4.23)$$

The FEC decoder then produces posterior LLRs $\mathbf{L}_{c_i}^d$ and will be used to form the decoder-aided soft symbol estimate for SIC cancellation. The calculation follows the same bit-probability, symbol-probability, and posterior-mean rules in equation (4.7), (4.8), and (4.9), with the a priori LLRs is replaced by the decoder posterior LLRs. Therefore, for each symbol position ℓ , the decoded soft estimate $\hat{s}_{i,\ell}^d$ is obtained from $\mathbf{L}_{c_i}^d$ in the same way that $\mu_{i,\ell}^{(t)}$ is obtained from $\mathbf{L}_{c_i,\ell}^{(t)}$.

The only additional quantity required in the decoder-aided SIC process is the cancellation residual variance $\bar{v}_i^{\text{can}}$, which is the second term of effective variance in equation (4.20). This quantity is computed according to $\hat{s}_{i,\ell}^d$ and follows the equation (4.9)–(4.12) with the soft symbol mean is replaced by the decoded soft estimate.

Summary of MMSE-SIC-IDD The overall procedure of the MMSE-SIC-IDD receiver is summarized in Algorithm 1.



Algorithm 1 MMSE-SIC-IDD receiver for one coded block

- 1: Initialize the a priori LLRs $L_{c_i, \ell, m}^{(0)} = 0$ for all users, coded-symbol positions, and bit positions.
 - 2: **for** $t = 0, 1, \dots, t_{\max}$ **do**
 - 3: Compute $\mu_{i, \ell}^{(t)}$ and $\sigma_{i, \ell}^{2(t)}$ for all users.
 - 4: Form the variance matrix $\mathbf{V}$ from the block-averaged variances.
 - 5: Determine the detection order and re-number the users as $1, \dots, U$.
 - 6: **for** $i = 1, 2, \dots, U$ **do**
 - 7: Compute the variance-aware MMSE filter in (4.14).
 - 8: Build $\mathbf{y}_{(i)}^{(t)}$ using (4.16) or (4.17).
 - 9: Generate detector extrinsic LLRs using (4.22).
 - 10: Form the decoder input LLRs using (4.23).
 - 11: Decode user i and obtain $\hat{s}_i^d$ and $\bar{v}_i^{can}$.
 - 12: **end for**
 - 13: Set the next a priori LLRs using (4.4).
 - 14: **end for**
 - 15: Make hard decisions from the final decoder outputs.
-

4.4 MMSE-PIC-IDD

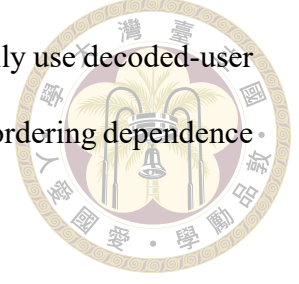
In this section, we introduce the MMSE-PIC-IDD receiver, which is a parallel interference cancellation (PIC) receiver that also exchanges soft information with the FEC decoder. Compared to the successive interference cancellation (SIC), PIC cancels all other users in parallel instead of successively decoding one user at a time [21]. For user k , the residual vector is calculated by subtracting all other users:

$$\mathbf{y}_{k, \ell}^{(t)} = \mathbf{y}_\ell - \sum_{j \neq k} \mathbf{h}_j \mu_{j, \ell}^{(t)}. \quad (4.24)$$

The variance-aware MMSE filter follows the same form as equation (4.14), and the LLR calculation follows the MMSE-SIC procedure.

PIC avoids SIC ordering and reduces decoding latency because users can be processed in parallel. However, it relies directly on the a priori information of all interfering users, so the iteration of a priori information update is especially important. SIC, in con-

trast, performs ordered stage-by-stage cancellation and can additionally use decoded-user feedback from earlier stages in the same pass, albeit at the expense of ordering dependence and increased latency.



4.5 MAP-IDD

In this receiver, the detector follows the maximum a posteriori (MAP) rule. The detector-decoder exchange follows the same IDD structure described previously. The detector produces bit-level LLRs, the FEC decoder refines them, and the decoder extrinsic information is fed back to the detector as the a priori information for the next update.

In this section, we mainly focus on how the a priori information is operated in the MAP detector. Similar to Section 2.3, the MAP detector considers all possible superimposed signals. For simplicity, the subscript ℓ is omitted in this section, which means that the receiver is processing the ℓ th transmission symbol. The possible superimposed signal set is written as

$$\mathcal{S} = \{\mathbf{s}_0, \mathbf{s}_1, \dots, \mathbf{s}_{N_L-1}\}, \quad (4.25)$$

where

$$N_L = \prod_{k=1}^U |\mathcal{S}_k| = 2^{\sum_{k=1}^U M_k}. \quad (4.26)$$

The ℓ^{th} possible superimposed signal is generated by one combination of all users' modulation symbols. Let this combination be

$$\mathbf{x}_l = [x_{1,l}, x_{2,l}, \dots, x_{U,l}]^T, \quad x_{k,l} \in \mathcal{S}_k. \quad (4.27)$$

Then the corresponding superimposed signal is

$$\mathbf{s}_l = \mathbf{H} \mathbf{x}_l. \quad (4.28)$$



For example, consider three users employing BPSK, QPSK, and 16QAM, respectively. In this case,

$$|\mathcal{S}_1| = 2, \quad |\mathcal{S}_2| = 4, \quad |\mathcal{S}_3| = 16,$$

and the MAP detector needs to consider

$$N_L = 2 \times 4 \times 16 = 128$$

possible superimposed signals for each received symbol. One possible combination is, for example:

$$\mathbf{x}_l = [x_{1,l}, x_{2,l}, x_{3,l}]^T = \left[+1, \frac{1+j}{\sqrt{2}}, \frac{1+3j}{\sqrt{10}} \right]^T, \quad (4.29)$$

where the first entry is selected from the BPSK constellation, the second entry is selected from the QPSK constellation, and the third entry is selected from the 16QAM constellation. This symbol combination produces one possible superimposed signal

$$\mathbf{s}_l = \mathbf{H} \left[+1, \frac{1+j}{\sqrt{2}}, \frac{1+3j}{\sqrt{10}} \right]^T. \quad (4.30)$$

The MAP detector repeats this construction for all possible symbol combinations.

Without a priori information, all possible superimposed signals are assumed to be equally likely. In MAP-IDD, however, the detector uses the a priori LLRs from the previous IDD update to weight each possible superimposed signal. From equation (4.7), the

a priori probability of the l th possible superimposed signal is

$$P_l^{a,(t)} = \prod_{k=1}^U \prod_{m=0}^{M_k-1} P_l^{a,(t)}(b_{k,m} = b_{k,m}(x_{k,l})). \quad (4.31)$$



Therefore, the APP of the l th possible superimposed signal is proportional to

$$P^{(t)}(\mathbf{s}_l|\mathbf{y}) \propto \exp\left(-\frac{\|\mathbf{y} - \mathbf{s}_l\|^2}{2\sigma^2}\right) P_l^{a,(t)}. \quad (4.32)$$

Equation (4.32) shows that the a priori information changes the probability of each possible superimposed signal according to its bit labels.

For the m^{th} bit of user k , the detector output LLR is obtained by summing the APPs of all possible superimposed signals whose corresponding bit is equal to 0 or 1:

$$L_{c_k,m}^{\text{MAP}} = \log \frac{\sum_{l:b_{k,m}(x_{k,l})=0} P^{(t)}(\mathbf{s}_l|\mathbf{y})}{\sum_{l:b_{k,m}(x_{k,l})=1} P^{(t)}(\mathbf{s}_l|\mathbf{y})}. \quad (4.33)$$

After the detector LLRs are derived synchronously for all users, the subsequent IDD procedure follows the same steps described previously.

MAP-IDD provides a strong coded joint-detection ability because it evaluates the APPs over all possible superimposed signals. However, its complexity increases with $N_L = \prod_{k=1}^U |\mathcal{S}_k|$, and therefore grows rapidly as the number of users or modulation order increases.

4.6 Coded Simulation Results

For the MMSE-SIC-IDD receiver, post-SINR ordering is used. In the four-user case and overloaded scenarios with more than four users, dynamic post-SINR ordering is con-

sidered. In this ordering strategy, the detection order is updated after each SIC stage. Accordingly, the post-detection SINR in equation (4.15) is recalculated at every SIC stage based on the remaining undecoded users.



We compare block error rate (BLER) of three different receiver structures that mentioned above. The simulations employ a (1008, 504) LDPC code over a independent block-fading channel for every user of each resource. The number of a priori information update iterations is denoted by t_{max} , where $t_{max} = 0$ means no a priori information update is used. 2-resources is used in our system and BPSK modulation for each user unless otherwise specified. $\bar{E}_b$ in the simulations represent the average transmit energy per bit per user, normalized by the number of resources.

The objective of these simulation results is to investigate the effects of decoder feedback, SIC/PIC-based interference cancellation, system loading, and modulation order on the performance of coded multi-user receivers.

Figure. 4.2 shows the BLER performance of MMSE-SIC-IDD and MAP-IDD with unequal- β_k case. Figure. 4.3 shows the BLER performance of MMSE-PIC-IDD with unequal- β_k case. It's observed that MMSE-PIC-IDD strongly relies on reliable a priori information update because parallel cancellation relies directly on the a priori-based soft means of all interfering users. Compare with MMSE-SIC-IDD and MAP-IDD in figure. 4.2, the performance is closed to all users, indicating that decoder-aided soft cancellation effectively reduces residual interference.

Figure. 4.4 shows the BLER performance of MMSE-PIC-IDD, MMSE-SIC-IDD and MAP-IDD with equal- β_k case. For the MMSE-PIC-IDD scheme, convergence of the BLER is nearly achieved by $t_{max} = 3$. Consequently, the remaining simulations in this

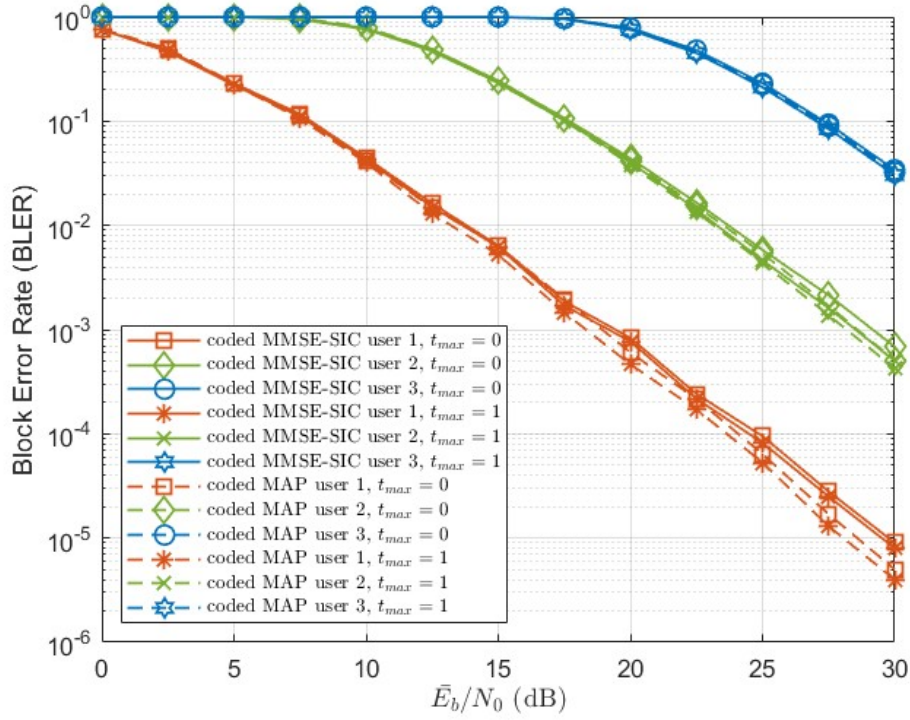
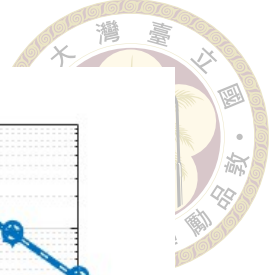


Figure 4.2: BLER of MMSE-SIC-IDD and MAP-IDD with $R = 2$, $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$, $t_{\max} = 0$ and $t_{\max} = 1$.

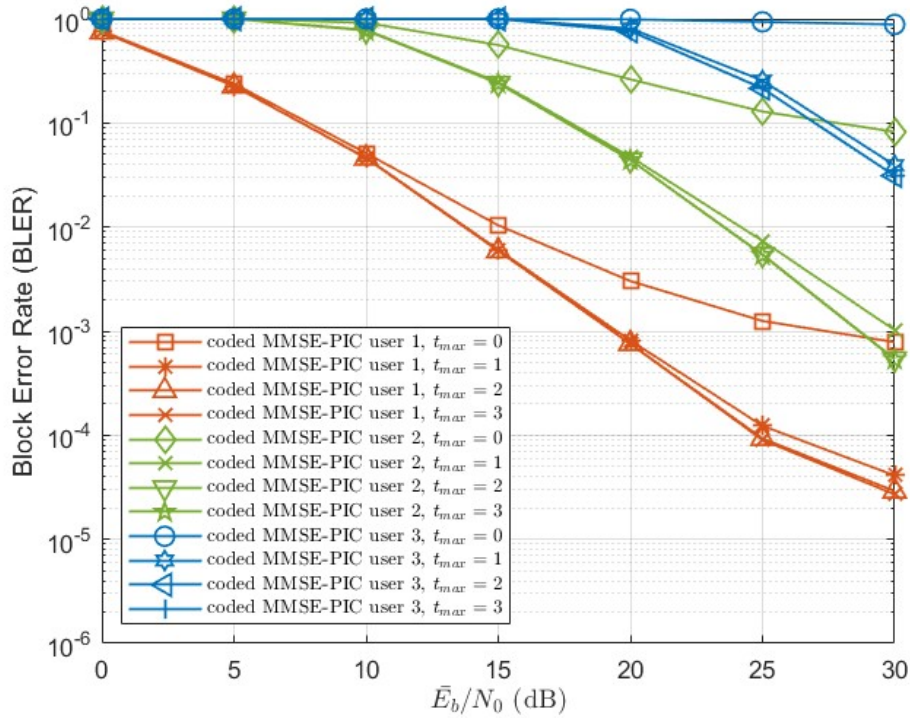


Figure 4.3: BLER of MMSE-PIC-IDD with $R = 2$, $(\beta_1, \beta_2, \beta_3) = (1, 0.1, 0.01)$, $t_{\max} = 0, 1, 2, 3$

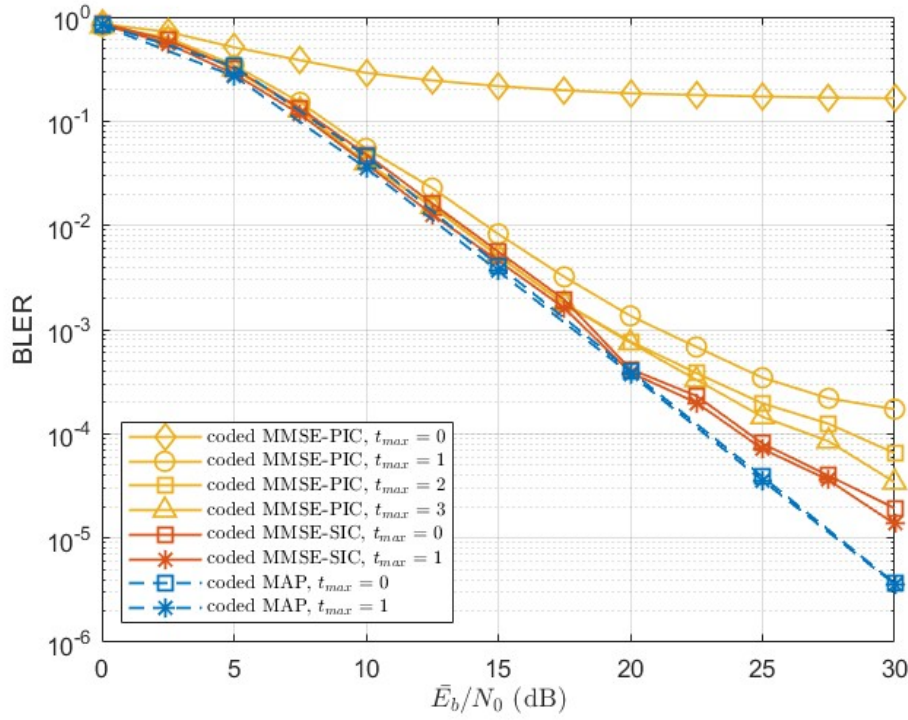


Figure 4.4: BLER of different receiver structures with $R = 2$, $U = 3$ and $(\beta_1, \beta_2, \beta_3) = (1, 1, 1)$.

chapter are conducted solely using $t_{\max} = 3$. Compared with the MMSE-SIC-IDD and MAP-IDD, the performance is also closed, while MAP-IDD is slightly better in the high-SNR region. It can also be observed that for both unequal- β_k case and equal- β_k case, the gain from increasing the value of t is limited. This is due to the assumption of a block fading channel and the relatively moderate system loading.

Figure. 4.5 and 4.6 are the case of 4-user and 6-user respectively. It can be seen that MAP-IDD maintains superior BLER performance owing to its near-optimal detection capability, whereas MMSE-based receivers becomes more sensitive to the increase in the number of user.

Figure. 4.7 and 4.8 consider three users using modulation (BPSK, QPSK, 16QAM). This two figures represent $t_{\max} = 0$ and $t_{\max} = 1$, respectively. Compare with BPSK modulation for every user, MAP-IDD can benefit from a priori information update even

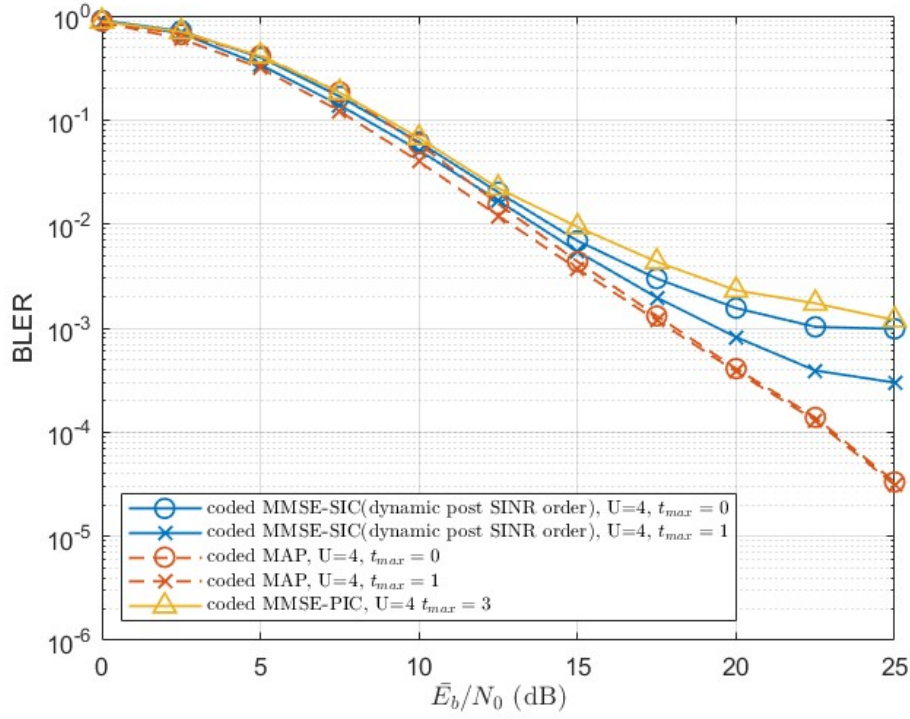
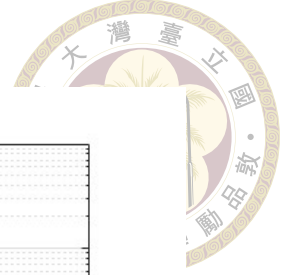


Figure 4.5: BLER of different receiver structures with $R = 2$, $U = 4$ and $\beta_k = 1$.

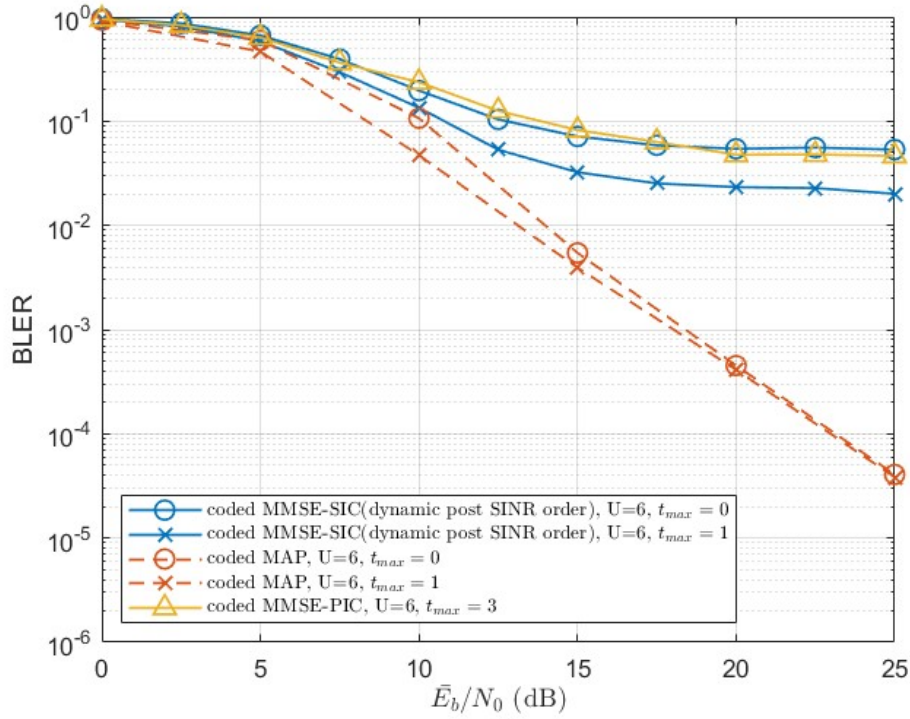


Figure 4.6: BLER of different receiver structures with $R = 2$, $U = 6$ and $\beta_k = 1$.

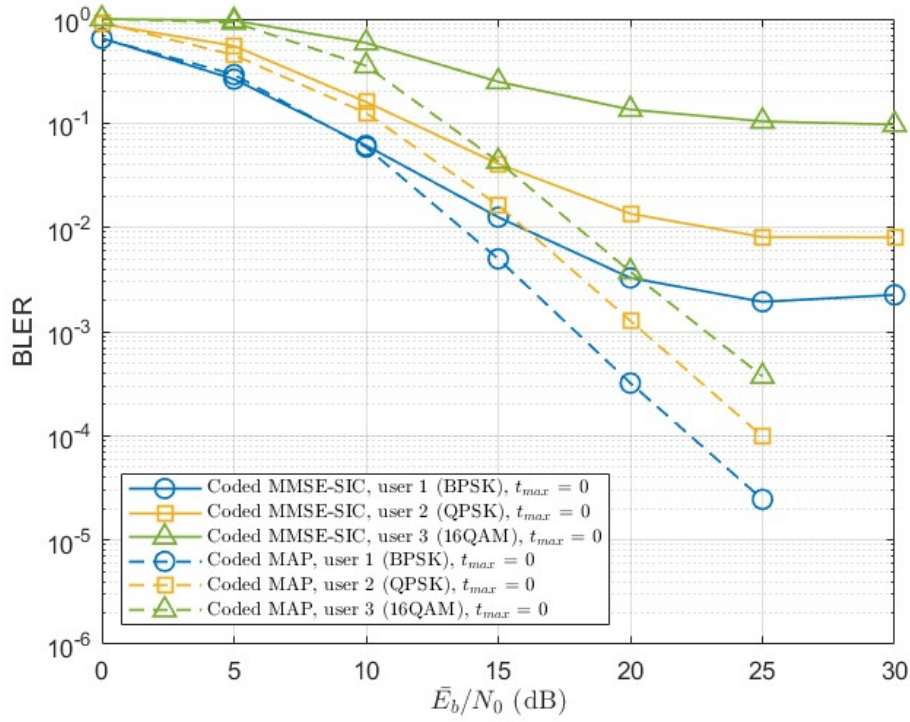
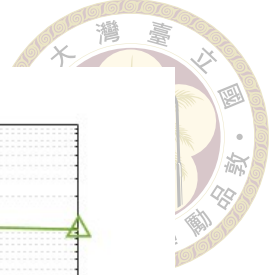


Figure 4.7: BLER of MMSE-SIC-IDD and MAP-IDD with $R = 2$, $U = 3$ and $\beta_k = 1$. (BPSK, QPSK, 16QAM) is used. $t_{\max} = 0$

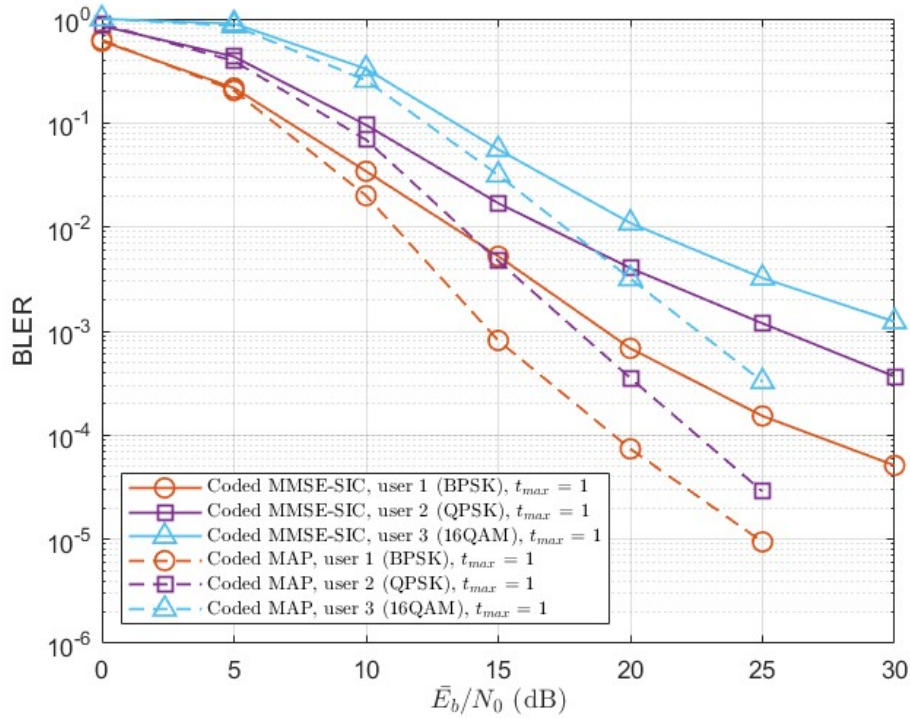
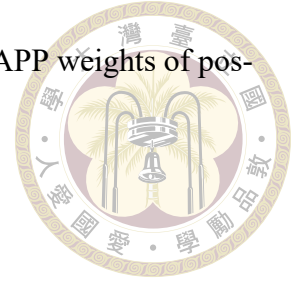


Figure 4.8: BLER of MMSE-SIC-IDD and MAP-IDD with $R = 2$, $U = 3$ and $\beta_k = 1$. (BPSK, QPSK, 16QAM) is used. $t_{\max} = 1$

under block fading channel, since decoder feedback helps refine the APP weights of possible superimposed signals.



4.7 Comparison and summary

The simulation results show that the IDD is effective for coded multi-resource NOMA receivers, but the amount of gain depends on the detector type, system loading, and modulation order. For MAP-IDD, the detector already provides highly reliable soft information. Therefore, the additional gain from the a priori information update is smaller than that of the MMSE-based receivers in simple BPSK cases. However, MAP-IDD still achieves the best BLER performance in all simulated scenarios and remains robust when the number of users or modulation order increases.

MMSE-SIC-IDD provides a favorable tradeoff between performance and complexity. By combining dynamic post-SINR ordering, variance-aware MMSE filtering, and decoder-aided cancellation, it can approach MAP-IDD in the three-user case and still performs well in the four-user case while BPSK modulation is used for every user. However, it is still based on linear filtering and sequential cancellation. When the system becomes heavily loaded, the residual interference causes a clear performance floor.

MMSE-PIC-IDD has the advantage of parallel processing and lower receiver latency because all users can be detected simultaneously. However, its cancellation quality strongly depends on the reliability of the a priori information.



Chapter 5 Coded OFDM NOMA system

This chapter extends the coded multi-resource receiver to an Orthogonal Frequency-Division Multiplexing (OFDM) system. Since practical broadband channels are frequency selective, OFDM is introduced to convert the multi-path channel into parallel frequency-domain observations. Furthermore, the high spectral efficiency of OFDM, together with its resilience to channel distortions, has led to its widespread adoption in practical systems such as LTE, Wi-Fi, and digital broadcasting standards.

We first evaluate the BLER performance of the synchronous coded OFDM NOMA system with different receivers discussed in Chapter 4. However, perfect synchronous transmission is impractical especially for uplink transmissions, where users may arrive at the base station with different timing offsets [7]. Therefore, this chapter further extends the study to asynchronous uplink transmission and examines the robustness of these receivers under user-dependent timing offsets.

5.1 OFDM System and channel Model



Related coded NOMA OFDM systems have been studied for robust multiuser transmission to enhance data rate[10, 14, 19]. Practical uplink systems often employ OFDM to handle frequency-selective fading and multipath interference. Consider a coded OFDM system with N_{FFT} subcarriers and cyclic prefix length N_{CP} . Let N_c denote the coded block length and M denote the number of coded bits per modulation symbol. Each coded block occupies

$$N_{\text{sym}} = \frac{N_c}{MN_{\text{FFT}}} \quad (5.1)$$

OFDM symbols.

After CP removal and FFT, the received signal on subcarrier q of OFDM symbol j is

$$\mathbf{y}_{q,j} = \mathbf{H}_q \mathbf{x}_{q,j} + \mathbf{n}_{q,j}. \quad (5.2)$$

where $q = 0, 1, \dots, N_{\text{FFT}} - 1$ and $j = 0, 1, \dots, N_{\text{sym}} - 1$. Here, $\mathbf{y}_{q,j} \in \mathbb{C}^{R \times 1}$ is the received vector over R resources, $\mathbf{x}_{q,j} \in \mathbb{C}^{U \times 1}$ is the transmitted symbol vector of U users, and $\mathbf{n}_{q,j} \in \mathbb{C}^{R \times 1}$ denotes complex AWGN. For each subcarrier, the OFDM system follows the same multi-resource vector model as (3.1).

The channel matrix on subcarrier q is written as

$$\mathbf{H}_q = \begin{bmatrix} \sqrt{P_1\beta_1}H_{1,1,q} & \sqrt{P_2\beta_2}H_{1,2,q} & \cdots & \sqrt{P_U\beta_U}H_{1,U,q} \\ \sqrt{P_1\beta_1}H_{2,1,q} & \sqrt{P_2\beta_2}H_{2,2,q} & \cdots & \sqrt{P_U\beta_U}H_{2,U,q} \\ \vdots & \vdots & \ddots & \vdots \\ \sqrt{P_1\beta_1}H_{R,1,q} & \sqrt{P_2\beta_2}H_{R,2,q} & \cdots & \sqrt{P_U\beta_U}H_{R,U,q} \end{bmatrix}, \quad (5.3)$$

where $H_{r,k,q}$ is the frequency-domain channel coefficient of user k on resource r and sub-carrier q .



Let $h_{r,k,p}$ denote the time-domain channel coefficient of user k on resource r at delay tap p , where $k = 1, 2, \dots, U$, $r = 1, 2, \dots, R$, and $p = 0, 1, \dots, L_h - 1$. The normalized power of the p th tap follows an exponentially decayed power-delay profile:

$$\rho_p = \frac{e^{-p}}{\sum_{p'=0}^{L_h-1} e^{-p'}}, \quad p = 0, 1, \dots, L_h - 1. \quad (5.4)$$

The first tap is modeled as a Rician fading path with a LOS component, while the remaining taps are modeled as Rayleigh fading paths [1]. Specifically, the first tap is generated as

$$h_{r,k,0} = \sqrt{\rho_0} \left(\sqrt{\frac{1}{\kappa + 1}} g_{r,k,0} + \sqrt{\frac{\kappa}{\kappa + 1}} e^{j\phi_{r,k}} \right), \quad (5.5)$$

and the remaining taps are generated as

$$h_{r,k,p} = \sqrt{\rho_p} g_{r,k,p}, \quad p = 1, 2, \dots, L_h - 1, \quad (5.6)$$

where κ is the Rician factor, $\phi_{r,k} \in [0, 2\pi)$ is the LOS phase and L_h is the number of channel taps. The NLOS component is modeled as

$$g_{r,k,p} = a_{r,k,p} + jb_{r,k,p}, \quad a_{r,k,p}, b_{r,k,p} \sim \mathcal{N}\left(0, \frac{1}{2}\right). \quad (5.7)$$

When $\kappa = 0$, the LOS component of first tap vanishes. The channel reduces to a Rayleigh fading model.

The corresponding frequency-domain channel coefficient on subcarrier q is obtained

by

$$H_{r,k,q} = \sum_{p=0}^{L_h-1} h_{r,k,p} e^{-j2\pi qp/N_{\text{FFT}}}. \quad (5.8)$$

Thus, the frequency-selective multipath channel is represented by different channel matrices $\mathbf{H}_q$ across subcarriers, while each subcarrier can still be processed using the multi-resource NOMA model in equation (5.2).

5.2 MMSE-SIC-IDD for OFDM

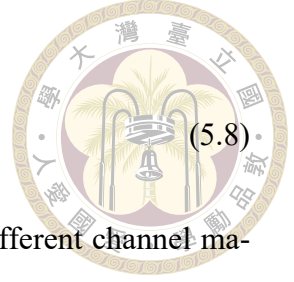
In practical wireless systems, the coded-OFDM is commonly implemented where the LLR on subcarriers are calculated and then sent to the decoder in a block manner[18]. Therefore, we calculate the post-filtering SINR(or the received-power metric) of each subcarrier q of user k and decide a detection order of whole coded block. The detection can be determined from an average reliability metric:

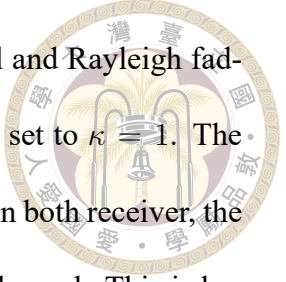
$$\bar{\Gamma}_k = \frac{1}{N_{\text{FFT}}} \sum_q \Gamma_{k,q}, \quad (5.9)$$

where $\Gamma_{k,q}$ follows the calculation in section 3.2.1.

5.3 Synchronous OFDM Simulation Results

We first compare MMSE-SIC-IDD, MMSE-PIC-IDD, and MAP-IDD under the synchronous OFDM system. The receivers follow the IDD procedures introduced in Chapter 4. Unless otherwise specified, the simulations consider a two-resource system with BPSK modulation for each user with $\beta_k = 1$. (1008, 504) LDPC code. The OFDM parameters are $N_{\text{FFT}} = 16$ and $N_{\text{CP}} = 4$.





We first compare a three-user system with Rician fading channel and Rayleigh fading channel of MMSE-SIC-IDD and MAP-IDD. The Rician factor is set to $\kappa = 1$. The simulation results are shown in Figure ?? and ?. It's can be seen that in both receiver, the Rayleigh fading channel has better performance than Rician fading channel. This is because the Rician fading channel has a strong LOS component, which reduces the frequency diversity of the channel. In contrast, the Rayleigh fading channel has more independent fading paths, which can provide better diversity gain and improve the performance of the receivers.

Figure. ?? shows the impact of the a priori information update on MMSE-PIC-IDD for the three-user case. When $t = 0$, the detector does not have updated decoder information, and the BLER performance is poor. As t increases, the decoder feedback provides more reliable soft information, and the BLER improves significantly.

Figure. ?? compares the two detection-order methods for MMSE-SIC-IDD. Both of them benefit from the a priori information update at high SNR region. However, post-SINR ordering provides better BLER performance because it accounts for the residual interference and the noise enhancement after MMSE filtering. Additionally, it requires 1 dB less SNR to achieve a BLER of 10^{-4} than MMSE-PIC-IDD with $t = 3$.

Figure. ?? compares MMSE-SIC-IDD and MAP-IDD for the three-user case. Since the loading is relatively light, MMSE-SIC-IDD with post-SINR ordering can achieve performance close to, and slightly better than, MAP-IDD in the simulated high-SNR region. This result shows that the combination of decoder-aided soft cancellation, a priori information update, and frequency-domain channel diversity can improve the user separation capability of MMSE-SIC-IDD. In contrast, MMSE-SIC-IDD with amplitude-based order-

ing is slightly worse than MAP-IDD at high SNR because its ordering metric is less related to the actual post-filtering reliability.

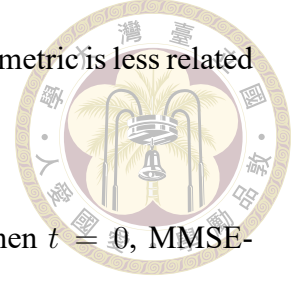
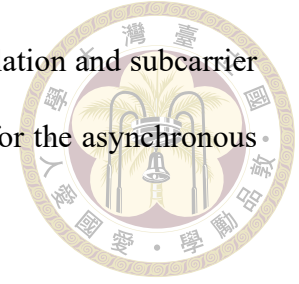


Figure. ?? and ?? show the effect of the number of users. When $t = 0$, MMSE-SIC-IDD becomes sensitive to error propagation as the number of users increases. This is because more users share the same resources, and the detector must cancel stronger multiuser interference through successive stages. If an early-stage decision or soft estimate is unreliable, the residual interference affects the following users and leads to BLER degradation. In contrast, MAP-IDD jointly evaluates the possible superimposed signals and therefore remains more stable when the number of users increases.

When $t = 1$, the decoder feedback improves the reliability of the soft estimates used in the SIC process, and MMSE-SIC-IDD obtains a clear performance gain. For the four-user case, MMSE-SIC-IDD can still approach MAP-IDD after one a priori update. However, when the number of users further increases, the performance gap becomes larger because the SIC process is still affected by residual cancellation errors and error propagation. Although additional a priori updates may further improve MMSE-SIC-IDD in frequency-selective channels, only $t = 0$ and $t = 1$ are mainly considered in the following comparisons to keep the receiver complexity practical and fair.

Overall, the synchronous OFDM results show that the IDD structure is effective for coded OFDM NOMA receivers. MMSE-PIC-IDD requires reliable a priori information because its parallel cancellation directly depends on the soft estimates of all users. MMSE-SIC-IDD is less dependent on full parallel soft cancellation and can achieve strong performance when a reliable detection order is used. MAP-IDD remains the most stable receiver as the number of users increases, but in lightly loaded cases, MMSE-SIC-IDD with

post-SINR ordering can be competitive due to decoder-aided cancellation and subcarrier channel diversity. These synchronous results serve as the baseline for the asynchronous OFDM study in the next section.



5.4 Asynchronous OFDM NOMA system

This section extends the coded OFDM NOMA system to asynchronous uplink transmission. In practical uplink transmissions, users may arrive at the BS with different timing offsets because of propagation delay, dynamic mobile environments, or limited timing-advance precision [2, 7, 15]. If the sum of the timing offset and the maximum channel delay spread remains within the cyclic prefix (CP), the OFDM subcarriers remain orthogonal and the timing offset only introduces a phase rotation in each subcarrier symbol [16]. Therefore, a timing offset within the CP can be represented as a subcarrier-dependent phase rotation, but the multiuser detector observes different effective channel responses for different users.

In most related studies, timing offset is treated as an impairment and the receiver attempts to estimate, compensate, or cancel it [7, 15, 16]. In contrast, timing offset can also be intentionally introduced into NOMA to decrease inter-user interference and enhance performance [12]. In this section, we evaluate the tolerance of the considered receivers when timing offsets exist in the received signals.

5.4.1 Asynchronous Signal Model

For a NOMA system, different users may arrive at the BS with different timing offsets. Therefore, the effective channel observed by the multiuser detector is changed, and

additional multiple-access interference (MAI) or inter-symbol interference (ISI) may appear depending on the receiver alignment and the CP margin. [15]. Let τ_k denote the timing offset of user k and set user 1 as the receiver timing reference, i.e., $\tau_1 = 0$. The relative timing offset of user k is defined as

$$\Delta_k = \tau_k - \tau_1. \quad (5.10)$$

For a multi-path OFDM channel, the timing offset can be represented only as a subcarrier-dependent phase rotation when the delayed channel impulse response is still fully contained within the CP [16]. Let $\tau_{\max}$ denote the maximum channel delay spread. In the five-tap channel model considered in this chapter, $\tau_{\max} = 4$. Therefore, the sufficient condition for preserving the circular convolution property of OFDM is

$$\Delta_k + \tau_{\max} \leq N_{\text{CP}}. \quad (5.11)$$

For user k of subcarrier q , the effective channel coefficient becomes

$$\tilde{\mathbf{H}}_{k,q} = \mathbf{H}_{k,q} e^{-j2\pi q \Delta_k / N_{\text{FFT}}}, \quad \tilde{\mathbf{H}}_{k,q} \in \mathbb{C}^{R \times 1} \quad (5.12)$$

Equivalently, for the multi-resource vector model, the asynchronous effective channel matrix on subcarrier q can be written as

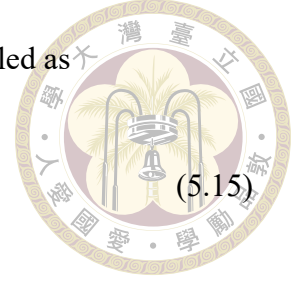
$$\tilde{\mathbf{H}}_q = \mathbf{H}_q \Phi_q, \quad \tilde{\mathbf{H}}_q \in \mathbb{C}^{R \times U} \quad (5.13)$$

where

$$\Phi_q = \text{diag} \left(e^{-j2\pi q \Delta_1 / N_{\text{FFT}}}, e^{-j2\pi q \Delta_2 / N_{\text{FFT}}}, \dots, e^{-j2\pi q \Delta_U / N_{\text{FFT}}} \right). \quad (5.14)$$

The received signal on subcarrier q of OFDM symbol j is then modeled as

$$\mathbf{y}_{q,j} = \tilde{\mathbf{H}}_q \mathbf{x}_{q,j} + \mathbf{n}_{q,j}. \quad (5.15)$$



The above model treats asynchronous transmission as a user-dependent effective channel variation. This model is exact when equation (5.11) holds. However, when the timing offset and the multipath delay spread exceed the CP margin, part of the delayed signal falls outside the FFT observation window. In this case, the received OFDM symbol can no longer be described only by the phase-rotated channel in equation (5.12). Additional ISI may appear. The multiuser detector may also observe stronger residual MAI due to the mismatch between the assumed frequency-domain model and the actual received signal.

In the asynchronous simulations, the detector still processes the received signal through the effective phase-rotated channel model in equation (5.13). Therefore, when equation (5.11) is violated, this model should be interpreted as a receiver-side approximation. The purpose of the section is to evaluate the robustness of MAP-IDD, MMSE-SIC-IDD, and MMSE-PIC-IDD under user-dependent timing offsets.

5.4.2 Asynchronous Simulation Results

The asynchronous simulations consider the two-resource systems with three-user and four-user. The receiver is aligned to user 1, and the relative timing offset of the other users is randomly selected from $[0, \Delta_{\max}]$. The CP length is $N_{\text{CP}} = 4$, and the channel follows the five-tap Rayleigh fading model used in the previous OFDM simulations. Therefore, $\Delta_{\max} = 4$ indicates that the timing offset itself is bounded by the CP length, while $\Delta_{\max} =$

5 represents a more severe asynchronous case. In our simulations, we consider $t = 0$ and $t = 1$ for MMSE-SIC-IDD and MAP-IDD, $t = 2$ for MMSE-PIC-IDD.

For the three-user case, Figure. ??-?? show the BLER of $\Delta_{max} = 4$. It can be seen that MMSE-SIC-IDD and MMSE-PIC-IDD remain competitive under asynchronous transmission. With $t = 0$, MAP-IDD achieves slightly better in high SNR region, while MMSE-SIC-IDD become better with $t = 1$ in simulated SNR range. This indicates that MMSE-based detection is relatively robust to asynchronous effects since they treat the remaining users as interference-plus-noise. Therefore, part of the asynchronous mismatch and residual multiuser interference can be absorbed into the effective interference-plus-noise term. In contrast, MAP-IDD relies more directly on the assumed likelihood model of the possible superimposed signals, so its advantage can be reduced when the phase-rotated frequency-domain model does not fully match the actual asynchronous received signal. The two MMSE-SIC ordering methods also achieve almost the same performance, indicating that the ordering gain becomes less significant when the dominant impairment comes from asynchronous system.

When $\Delta_{max} = 5$, the asynchronous effect becomes stronger. A similar trend can be observed compared with $\Delta_{max} = 4$. In particular, Figure. ?? shows that even without a priori information update, the two MMSE-SIC-IDD detection-order strategies can outperform MAP-IDD.

For the four-user case, figure. ?? shows that with $t = 0$, MAP-IDD performs better than MMSE-SIC-IDD since the residual interference becomes severe when the number of users increases. In the synchronous four-user system, MAP detection usually has a clearer advantage because it jointly evaluates the possible superimposed signals. How-

ever, under asynchronous transmission, this advantage becomes less pronounced. With $t = 1$, MMSE-SIC-IDD can achieve BLER performance slightly better than MAP-IDD. This result suggests that the interference-plus-noise modeling in MMSE-SIC-IDD provides additional tolerance to asynchronous mismatch.

The case of $\Delta_{\max} = 5$ follows a similar trend with $\Delta_{\max} = 4$ while the overall BLER become worse. Under this condition, the performance is dominated not only by multiuser detection capability but also by the receiver's tolerance to residual MAI/ISI-like effects.

Overall, the asynchronous simulation results show that MMSE-SIC-IDD is more robust to asynchronous transmission than expected from the synchronous comparison. The two detection-order strategies of MMSE-SIC-IDD also show almost identical BLER performance, especially when $\Delta_{\max} = 5$. This implies that, in the presence of stronger asynchronous interference, the receiver performance is dominated more by the interference-plus-noise modeling and residual cancellation behavior than by the fine difference between the ordering metrics.



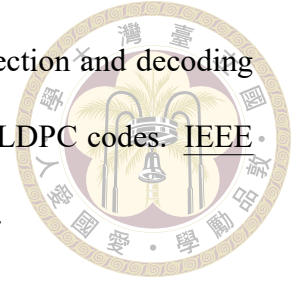
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Appendix A — Introduction

A.1 Introduction

A.2 Further Introduction



Appendix B — Introduction

B.1 Introduction

B.2 Further Introduction